

Unit 3. Bases, Dimension, and Rank

1. Bases and the number of independent vectors

A **basis** of a real linear space V is a finite list $v_1, \dots, v_r$ that is linearly independent and spans V . The empty list is a basis of $\{0\}$: its only linear combination is zero, and it has no nonzero coefficient relation. For a basis, every $v \in V$ has exactly one expression

$$v = c_1 v_1 + \dots + c_r v_r.$$

Existence is the spanning property. Subtracting two expressions gives $\sum_i (c_i - d_i) v_i = 0$; independence gives $c_i = d_i$ for all i . The standard columns $e_1, \dots, e_n$ form a basis of $\mathbb{R}^n$, since $x = \sum_i x_i e_i$ and coordinate i of $\sum_j c_j e_j$ is c_i .

Example 3.1. Show that $u = (1, 1)^\top$ and $v = (-1, 2)^\top$ form a basis of $\mathbb{R}^2$, and express an arbitrary $(a, b)^\top$ in this basis.

The coefficient equations are

$$x - y = a, \quad x + 2y = b.$$

Subtracting the first from the second gives $3y = b - a$, so

$$\begin{pmatrix} a \\ b \end{pmatrix} = \frac{2a+b}{3} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{b-a}{3} \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

These coefficients exist for every a, b . For $a = b = 0$ both are zero, which proves independence as well as spanning.

Proposition 3.1 (Adjoining one vector). If $v_1, \dots, v_r$ are independent, then

$$v_1, \dots, v_r, w \text{ are independent} \iff w \notin \text{span}(v_1, \dots, v_r).$$

Proof. • If $w = \sum_i a_i v_i$, then $\sum_i a_i v_i - w = 0$ has coefficient -1 on w , so the enlarged list is dependent.

- Suppose $w \notin \text{span}(v_1, \dots, v_r)$ and $\sum_i c_i v_i + dw = 0$. If $d \neq 0$, then $w = -\sum_i (c_i/d) v_i$, a contradiction. Thus $d = 0$; independence of the original list gives every $c_i = 0$.

□

Theorem 3.2 (Independent lists and spanning lists). If $v_1, \dots, v_m$ span V , every independent list in V has at most m members.

Proof. • If $m = 0$, then $V = \{0\}$, so every nonempty list is dependent.

- Suppose $m > 0$ and $w_1, \dots, w_k \in V$, with $k > m$. Write their coefficients in columns of an $m \times k$ matrix $C = (c_{ij})$:

$$w_j = \sum_{i=1}^m c_{ij} v_i \quad (1 \leq j \leq k).$$

- By Unit 1, Proposition 1.7, $Ct = 0$ has a nonzero solution $t \in \mathbb{R}^k$. Regrouping the finite sum gives

$$\sum_{j=1}^k t_j w_j = \sum_{i=1}^m \left(\sum_{j=1}^k c_{ij} t_j \right) v_i = 0.$$

Thus the w_j are dependent.

□

Theorem 3.3 (Extracting and extending a basis). Let V be spanned by a finite list.

- (a) A basis can be extracted from any finite spanning list.
- (b) Any independent list in V can be extended to a basis of V .
- (c) Every subspace $W \subseteq V$ has a finite basis.

Proof. • Scan the spanning list from left to right, retaining a vector exactly when it lies outside the span of those already retained. Proposition 3.1 preserves independence at each step. Every discarded vector belongs to the final span, so the retained list still spans V .

- To retain specified independent vectors, place them first, followed by a spanning list of V , and apply the same procedure. None of the specified vectors is discarded.
- Starting with the empty list, choose a vector of W outside the current span whenever one exists. The list stays independent. If V is spanned by m vectors, Theorem 3.2 forces the procedure to stop after at most m choices. At that point the retained list spans W . If $W = \{0\}$, it stops without choosing a vector.

□

2. Dimension and adapted bases

Any two bases of V have the same number of members: if their lengths are r, s , Theorem 3.2 gives both $r \leq s$ and $s \leq r$. This common number is the **dimension**, denoted $\dim V$. In particular $\dim\{0\} = 0$ and $\dim \mathbb{R}^n = n$.

Let E_{ij} be the $m \times n$ matrix with its only nonzero entry, equal to 1, in position (i, j) . Then

$$A = \sum_{i=1}^m \sum_{j=1}^n a_{ij} E_{ij}.$$

Reading entry (i, j) proves uniqueness, so the mn matrices E_{ij} form a basis and $\dim \mathbb{R}^{m \times n} = mn$. In the polynomial coefficient space $\mathcal{P}_d$ of Unit 2, $1, t, \dots, t^d$ form a basis by the same coefficient argument; thus $\dim \mathcal{P}_d = d + 1$.

Proposition 3.4 (Dimension tests). Let $\dim V = n$ and $W \subseteq V$ be a subspace.

- (a) Every independent list of n vectors in V is a basis.
- (b) Every spanning list of n vectors in V is a basis.
- (c) $\dim W \leq n$, with equality exactly when $W = V$.

Proof. • Extend the independent list to a basis. Every basis has n members, so no vector can have been added.

- Extract a basis from the spanning list. Its length is n , so no vector can have been discarded.
- A basis of W is independent in V , giving the inequality. At equality it is a basis of V by (a), and hence $W = V$. The reverse implication follows from equality of the spaces.

□

Example 3.2. Let V be the column span of

$$A = \begin{pmatrix} 1 & 2 & 3 \\ -2 & 3 & 8 \\ 5 & 1 & -3 \\ -3 & -4 & -5 \end{pmatrix} = [a_1 \ a_2 \ a_3].$$

Find a basis of V , its dimension, and an extension to a basis of $\mathbb{R}^4$.

The relation $a_3 = 2a_2 - a_1$ permits deletion of a_3 . If $sa_1 + ta_2 = 0$, the first two coordinates give

$$s + 2t = 0, \quad -2s + 3t = 0 \implies 7t = 0, s = 0.$$

Thus (a_1, a_2) is a basis of V and $\dim V = 2$. For an extension, use (a_1, a_2, e_3, e_4) . In a zero combination of these four columns the first two coordinates again force the coefficients of a_1, a_2 to be zero; coordinates 3 and 4 then force the other two coefficients to be zero. Four independent columns in $\mathbb{R}^4$ form a basis.

Theorem 3.5 (Dimension of a sum). For subspaces U, W of a finite-dimensional real space,

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Proof. • Choose a basis $c_1, \dots, c_k$ of $U \cap W$, and extend it to bases

$$(c_1, \dots, c_k, u_1, \dots, u_p) \text{ of } U, \quad (c_1, \dots, c_k, w_1, \dots, w_q) \text{ of } W.$$

- The combined list $(c_1, \dots, c_k, u_1, \dots, u_p, w_1, \dots, w_q)$ spans $U + W$: add the two basis representations and collect the c_i coefficients.
- For independence, suppose

$$\sum_i a_i c_i + \sum_j b_j u_j + \sum_\ell d_\ell w_\ell = 0.$$

Then

$$\sum_\ell d_\ell w_\ell = -\sum_i a_i c_i - \sum_j b_j u_j \in U \cap W = \text{span}(c_1, \dots, c_k).$$

Write this vector as $\sum_i e_i c_i$. The relation $\sum_\ell d_\ell w_\ell - \sum_i e_i c_i = 0$ in the basis of W gives every $d_\ell = e_i = 0$. The original relation now lies in the basis of U , giving every $a_i = b_j = 0$.

- The combined basis has $k + p + q = (k + p) + (k + q) - k$ members.

□

Since $U + W \subseteq V$, this also gives

$$\dim(U \cap W) \geq \dim U + \dim W - \dim V.$$

In particular, if $\dim U + \dim W > \dim V$, the intersection contains a nonzero vector. For $U \cap W = \{0\}$, the formula becomes $\dim(U \oplus W) = \dim U + \dim W$.

Example 3.3. In $\mathbb{R}^{n \times n}$, find bases and dimensions for the upper triangular space T , the symmetric space S , and $T \cap S$; determine $T + S$.

A basis of T is E_{ij} , $i \leq j$, since an upper triangular matrix has the unique expression $\sum_{i \leq j} a_{ij} E_{ij}$. A symmetric matrix has the unique expression

$$\sum_i a_{ii} E_{ii} + \sum_{i < j} a_{ij} (E_{ij} + E_{ji}).$$

Hence a basis of S consists of E_{ii} and $E_{ij} + E_{ji}$, $i < j$. There are n diagonal positions and $n(n-1)/2$ unordered distinct pairs: each of the $n(n-1)$ ordered pairs belongs to one pair of reversals. Thus $\dim T = \dim S = n(n+1)/2$. An upper triangular symmetric matrix is diagonal, so

$$T \cap S = \text{span}(E_{11}, \dots, E_{nn}), \quad \dim(T \cap S) = n.$$

The sum formula gives $\dim(T + S) = n^2$. Since $T + S \subseteq \mathbb{R}^{n \times n}$, Proposition 3.4 gives $T + S = \mathbb{R}^{n \times n}$.

Proposition 3.6 (Complements). Every subspace $W \subseteq V$ of a finite-dimensional space has a subspace U such that $V = W \oplus U$.

Proof. • Extend a basis $(w_1, \dots, w_r)$ of W to a basis $(w_1, \dots, w_r, u_1, \dots, u_s)$ of V , and set $U = \text{span}(u_1, \dots, u_s)$.

- The extended basis spans V , so $W + U = V$.
- If $\sum_i a_i w_i = \sum_j b_j u_j$, independence of the full list gives all $a_i = b_j = 0$. Thus $W \cap U = \{0\}$, proving the direct sum.

□

Example 3.4. Let $W = \text{span}((2, 1)^\top)$ in $\mathbb{R}^2$. Show that both $U = \text{span}((0, 1)^\top)$ and $U' = \text{span}((1, 1)^\top)$ are complements of W .

For every $(x, y)^\top$,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{x}{2} \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \left(y - \frac{x}{2}\right) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = (x - y) \begin{pmatrix} 2 \\ 1 \end{pmatrix} + (2y - x) \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The equations $a(2, 1)^\top = b(0, 1)^\top$ and $a(2, 1)^\top = b(1, 1)^\top$ each force $a = b = 0$, by their two coordinates. Both sums are therefore direct. Since $(0, 1)^\top \notin U'$, the two complements differ.

3. Rank and rank factorization

For $A \in \mathbb{R}^{m \times n}$, let $\mathcal{C}(A)$ be its column space, $N(A) = \{x : Ax = 0\}$, and define its **row space**, represented by columns, as $\mathcal{R}(A) = \mathcal{C}(A^\top) \subseteq \mathbb{R}^n$. The column rank is $\dim \mathcal{C}(A)$; the row rank is $\dim \mathcal{R}(A)$.

Theorem 3.7 (Equality of the two ranks). The column rank and row rank of every real matrix are equal. Their common value is denoted $\text{rank } A$.

Proof. • If $A = 0$, both spaces are $\{0\}$, so both ranks are zero.

- For $A \neq 0$, let $c = \dim \mathcal{C}(A)$. Extract a basis $b_1, \dots, b_c$ from the columns of A and put $B = [b_1 \ \dots \ b_c]$. Express each column of A in this basis and collect the coefficient columns in F :

$$A = BF, \quad B : m \times c, \quad F : c \times n.$$

- Row i of A is $\sum_{j=1}^c b_{ij} \text{row}_j(F)$. Hence $\mathcal{R}(A) \subseteq \mathcal{R}(F)$, and

$$\dim \mathcal{R}(A) \leq c = \dim \mathcal{C}(A).$$

- Applying the same inequality to $A^\top$ reverses it:

$$\dim \mathcal{C}(A) = \dim \mathcal{R}(A^\top) \leq \dim \mathcal{C}(A^\top) = \dim \mathcal{R}(A).$$

Thus the two ranks are equal.

□

Consequently,

$$0 \leq \text{rank } A \leq \min(m, n), \quad \text{rank } A^\top = \text{rank } A, \quad \text{rank } A = 0 \iff A = 0.$$

The bound follows because the column and row spaces lie in $\mathbb{R}^m$ and $\mathbb{R}^n$. Rank zero means every column belongs to $\{0\}$. A matrix has **full column rank** if its n columns are independent, equivalently $\text{rank } A = n$; it has **full row rank** if $\text{rank } A = m$. These equivalences follow by extracting bases from the respective lists.

Proposition 3.8 (Full rank factors). If $\text{rank } A = r > 0$, there is a factorization

$$A = BF, \quad B \in \mathbb{R}^{m \times r}, \quad F \in \mathbb{R}^{r \times n}, \quad \text{rank } B = \text{rank } F = r.$$

For this factorization,

$$\mathcal{C}(A) = \mathcal{C}(B), \quad \mathcal{R}(A) = \mathcal{R}(F), \quad N(A) = N(F).$$

Proof. • Use the extracted-column construction in Theorem 3.7. The basis columns give $\text{rank } B = r$ and $\mathcal{C}(A) = \mathcal{C}(B)$.

- At the r selected column positions, the corresponding columns of F are $e_1, \dots, e_r$, by uniqueness of basis coefficients. Thus $\text{rank } F = r$.
- Since $\mathcal{R}(A) \subseteq \mathcal{R}(F)$ and both have dimension r , they are equal.
- Finally $BFx = 0$ implies $Fx = 0$ by independence of the columns of B ; the converse follows by multiplication by B .

□

This is a **rank factorization**. For $r = 0$, it is the product of an $m \times 0$ and a $0 \times n$ matrix: each entry is an empty sum, hence zero.

Proposition 3.9 (Invertible multiplication). If P and Q are invertible of orders m and n , then

$$\text{rank}(PAQ) = \text{rank } A.$$

Proof. • The rows of PA are combinations of rows of A ; $A = P^{-1}(PA)$ gives the reverse inclusion. Therefore $\mathcal{R}(PA) = \mathcal{R}(A)$ and $\text{rank}(PA) = \text{rank } A$.

- The columns of AQ are combinations of columns of A ; $A = (AQ)Q^{-1}$ gives the reverse inclusion. Therefore $\mathcal{C}(AQ) = \mathcal{C}(A)$ and $\text{rank}(AQ) = \text{rank } A$.
- Apply both equalities: $\text{rank}(PAQ) = \text{rank}(AQ) = \text{rank } A$.

□

4. Computing bases and all solutions

Theorem 3.10 (The bases revealed by elimination). Suppose row reduction gives an echelon matrix $H = EA$, where E is invertible, with nonzero rows $1, \dots, r$ and pivot columns $j_1 < \dots < j_r$.

- The r nonzero rows of H , transposed, are a basis of $\mathcal{R}(A)$.
- The original columns $a_{j_1}, \dots, a_{j_r}$ are a basis of $\mathcal{C}(A)$.
- In particular, $\text{rank } A = r$.

Proof. • The nonzero rows span the row space of H , which equals $\mathcal{R}(A)$ by the proof of Proposition 3.9. In a zero combination of these rows, inspecting pivot position j_1 forces the first coefficient to be zero. Inspecting $j_2, j_3, \dots$ in order forces all remaining coefficients to be zero. Thus the row rank is r .

- For the pivot columns of H , inspect row r , then row $r - 1$, and continue upward in a zero combination. Each pivot is nonzero, so all coefficients vanish. There are $r = \text{rank } H$ independent pivot columns; they form a basis of $\mathcal{C}(H)$.

- Multiplication by E preserves every coefficient relation among columns (Unit 2, Proposition 2.5). Thus the original pivot columns are independent. If $h_j = \sum_i c_i h_{j_i}$, then

$$E \left(a_j - \sum_i c_i a_{j_i} \right) = 0 \implies a_j = \sum_i c_i a_{j_i}.$$

They also span $\mathcal{C}(A)$, so they form a basis.

□

Starting at the last pivot, divide its row by the pivot and clear all entries above it; continue upward. In each row being used, earlier pivot columns are zero by echelon form, and later pivot columns are zero from the preceding clearing steps. Thus all other pivot columns are preserved. The resulting matrix R has a 1 in each pivot position and zeros everywhere else in a pivot column; it is a **reduced echelon matrix**. Its zero rows may be omitted when recording equations. This construction does not assert uniqueness.

Theorem 3.11 (Null-space basis). Let R be a reduced echelon matrix for A , with pivots $j_1, \dots, j_r$ and free column set F . For each $k \in F$, set

$$z_k = e_k - \sum_{i=1}^r R_{ik} e_{j_i}.$$

Then $(z_k)_{k \in F}$ is a basis of $N(A)$. In particular,

$$\dim N(A) = n - r, \quad \text{rank } A + \dim N(A) = n.$$

Proof. • In row i , the equation $Rx = 0$ reads

$$x_{j_i} + \sum_{k \in F} R_{ik} x_k = 0.$$

Thus every solution has the expression $x = \sum_{k \in F} x_k z_k$, and conversely each such expression satisfies every row.

- The free coordinate $\ell \in F$ of z_k is 1 if $\ell = k$ and 0 otherwise, so a zero combination of the z_k has all coefficients zero.
- Row operations preserve solutions, hence this is a basis of $N(A)$. There are $n - r$ free columns. When F is empty, the basis is empty and $N(A) = \{0\}$; for $A = 0$, it is the standard basis of $\mathbb{R}^n$.

□

Example 3.5. For

$$A = \begin{pmatrix} 1 & 5 & 6 \\ 2 & 6 & 8 \\ 7 & 1 & 8 \end{pmatrix},$$

find its rank, bases for its column, row, and null spaces, a rank factorization, and all solutions of $Ax = b$ for an arbitrary $b = (b_1, b_2, b_3)^\top$.

Apply $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - 7R_1$, then $R_3 \leftarrow R_3 - \frac{17}{2}R_2$:

$$A \longrightarrow \begin{pmatrix} 1 & 5 & 6 \\ 0 & -4 & -4 \\ 0 & -34 & -34 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 5 & 6 \\ 0 & -4 & -4 \\ 0 & 0 & 0 \end{pmatrix} \longrightarrow R = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

The last step divides row 2 by -4 and subtracts five times that row from row 1. There are two pivots, in columns 1 and 2. Therefore

$$\text{rank } A = 2, \quad \mathcal{C}(A) = \text{span} \left(\begin{pmatrix} 1 \\ 2 \\ 7 \end{pmatrix}, \begin{pmatrix} 5 \\ 6 \\ 1 \end{pmatrix} \right), \quad \mathcal{R}(A) = \text{span} \left(\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right).$$

Each displayed list is a basis. The free coordinate is $x_3 = t$, giving

$$N(A) = \text{span} \left(\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \right), \quad A = \begin{pmatrix} 1 & 5 \\ 2 & 6 \\ 7 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

The product reproduces each original column, including $a_3 = a_1 + a_2$.

Applying the same row operations to the right-hand side gives

$$[A \mid b] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & (5b_2 - 6b_1)/4 \\ 0 & 1 & 1 & (2b_1 - b_2)/4 \\ 0 & 0 & 0 & b_3 + 10b_1 - 17b_2/2 \end{array} \right].$$

Thus the system is consistent exactly when $20b_1 - 17b_2 + 2b_3 = 0$, and then its full solution set is

$$x = \begin{pmatrix} (5b_2 - 6b_1)/4 \\ (2b_1 - b_2)/4 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

Proposition 3.12 (Rank tests for solutions). For $A \in \mathbb{R}^{m \times n}$,

- (a) $Ax = b$ is consistent exactly when $\text{rank}[A \mid b] = \text{rank } A$.
- (b) A consistent system has a unique solution exactly when $\text{rank } A = n$.
- (c) $Ax = b$ is consistent for every $b \in \mathbb{R}^m$ exactly when $\text{rank } A = m$.

Proof. • The inclusion $\mathcal{C}(A) \subseteq \mathcal{C}([A \mid b])$ is equality exactly when $b \in \mathcal{C}(A)$. For nested subspaces, equality is equivalent to equality of dimensions.

- By Unit 1, Theorem 1.9, every nonempty solution set is $x_0 + N(A)$. It is a singleton exactly when $\dim N(A) = 0$, equivalently $\text{rank } A = n$ by Theorem 3.11.
- Solvability for every b says $\mathcal{C}(A) = \mathbb{R}^m$, equivalently $\dim \mathcal{C}(A) = m$.

□

Exercises and solutions

Suggested practice: Exercises 3.1, 3.9, 3.7, 3.12, 3.23, and 3.26.

Bases, dimensions, and complements

Exercise 3.1 (Selecting columns). Let $a_1, \dots, a_9$ be the columns of

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 5 \\ 0 & 1 & 1 & 2 & 2 & 3 & -1 & 3 & 3 \\ 1 & 1 & 2 & 3 & 5 & 5 & 1 & 0 & 4 \end{pmatrix}.$$

Which lists are bases of $\mathbb{R}^3$?

- (a) (a_1, a_2) .
- (b) (a_1, a_4, a_6, a_8) .
- (c) (a_2, a_4, a_7) .
- (d) (a_3, a_5, a_9) .

Solution. (a) and (b) have the wrong lengths, since every basis of $\mathbb{R}^3$ has three members. For (c), a zero combination gives

$$s + t + 2u = 0, \quad s + 2t - u = 0, \quad s + 3t + u = 0.$$

Successive differences give $t - 3u = 0$ and $t + 2u = 0$; hence $u = t = s = 0$. Three independent vectors form a basis of $\mathbb{R}^3$. For (d), the nonzero coefficient relation $7a_3 - 2a_5 - a_9 = 0$ proves dependence.

Exercise 3.2 (Combining a given basis). If (x, y, z) is a basis of $\mathbb{R}^3$, which lists are also bases?

- (a) $(x + 2y, y + 3z, x + 2z)$.
- (b) $(x + y - 2z, x - 2y + z, -2x + y + z)$.
- (c) $(x, y, x + y + z)$.

Solution. For (a), a zero combination with coefficients a, b, c gives

$$(a + c)x + (2a + b)y + (3b + 2c)z = 0.$$

Thus $c = -a$, $b = -2a$, and $-8a = 0$; all coefficients vanish. The list is a basis. In (b) the three displayed vectors sum to zero, so they are dependent and not a basis. For (c),

$$ax + by + c(x + y + z) = (a + c)x + (b + c)y + cz = 0$$

forces $c = a = b = 0$; the list is a basis.

Exercise 3.3 (Mixing two bases). Suppose (x_1, x_2) and (y_1, y_2) are bases of $\mathbb{R}^2$. Prove that at least one of the following statements holds:

- (a) Both (x_1, y_2) and (x_2, y_1) are bases.
- (b) Both (x_1, y_1) and (x_2, y_2) are bases.

Solution. Write $y_1 = ax_1 + bx_2$, $y_2 = cx_1 + dx_2$. Comparison of coefficients in the x -basis shows

$$(x_1, y_2) \text{ is independent} \iff d \neq 0, \quad (x_2, y_1) \text{ is independent} \iff a \neq 0.$$

The analogous conditions for the two pairs in (b) are $b \neq 0$ and $c \neq 0$. If $a, d \neq 0$, (a) holds. If $a = 0$, then $b \neq 0$ since $y_1 \neq 0$; also $c \neq 0$, since otherwise both y_1, y_2 would lie in $\text{span}(x_2)$. If $d = 0$, then $c \neq 0$ and the same argument forces $b \neq 0$. Thus whenever (a) fails, (b) holds.

Exercise 3.4 (Spaces given by equations). Verify that each set is a subspace and find a basis and its dimension:

(a) $\{(a, b, c, d)^\top : a + b = c + d\} \subseteq \mathbb{R}^4$.

(b) $\{(x, y, z)^\top : 2x - y = 0, 2y + 3z = 0\} \subseteq \mathbb{R}^3$.

Solution. Both are solution sets of homogeneous linear systems, hence subspaces by Unit 1, Proposition 1.8. In (a), solve $a = -b + c + d$, obtaining exactly

$$b \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + c \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + d \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Coordinates 2, 3, and 4 establish independence, so the dimension is 3. In (b), $y = 2x$ and $z = -4x/3$; the set is $\{t(3, 6, -4)^\top : t \in \mathbb{R}\}$. The displayed nonzero column is a basis, and the dimension is 1.

Exercise 3.5 (Zero-sum columns). For $n \geq 1$, find a basis of $W = \{x \in \mathbb{R}^n : \sum_{i=1}^n x_i = 0\}$ and verify that it is a subspace.

Solution. Zero belongs to W , and the coordinate sum of $sx + ty$ is $s \sum_i x_i + t \sum_i y_i = 0$ for $x, y \in W$. Solving for the last coordinate gives

$$x = \sum_{i=1}^{n-1} x_i(e_i - e_n).$$

The first $n-1$ coordinates of a zero combination force all coefficients to vanish. Thus $(e_1 - e_n, \dots, e_{n-1} - e_n)$ is a basis and $\dim W = n - 1$. For $n = 1$, this is the empty basis of $\{0\}$.

Exercise 3.6 (Trace-zero matrices). Determine the dimension of $W = \{A \in \mathbb{R}^{n \times n} : \operatorname{tr} A = 0\}$, where $n \geq 1$, by exhibiting a basis.

Solution. Linearity of trace proves that W is a subspace. A basis is

$$\{E_{ij} : i \neq j\} \cup \{E_{ii} - E_{nn} : 1 \leq i < n\}.$$

Indeed, $a_{nn} = -\sum_{i < n} a_{ii}$, so every $A \in W$ equals

$$\sum_{i \neq j} a_{ij} E_{ij} + \sum_{i < n} a_{ii} (E_{ii} - E_{nn}).$$

Reading each off-diagonal entry and then diagonal entries $1, \dots, n-1$ proves independence. The dimension is $n(n-1) + (n-1) = n^2 - 1$. For $n = 1$, $W = \{0\}$, consistently with the empty basis.

Exercise 3.7 (Equal row and column sums). Find bases for the spaces of (a) 2×2 and (b) 3×3 real matrices whose row sums and column sums are all equal. The common sum may depend on the matrix.

Solution. The defining equalities are homogeneous linear equations in the entries, so both sets are subspaces. For (a), comparison of the two row sums with the column sums gives precisely

$$\begin{pmatrix} a & b \\ b & a \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The two coefficient matrices form a basis: their first rows distinguish the coefficients. The dimension is 2.

For (b), write the top left 2×2 block as $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, and let the bottom right entry be $-e$. If the common sum is s , the remaining entries in the first two rows and columns are

$$a_{13} = s - a - b, \quad a_{23} = s - c - d, \quad a_{31} = s - a - c, \quad a_{32} = s - b - d.$$

The last column sum gives $s = a + b + c + d + e$. Thus the complete family is

$$\begin{pmatrix} a & b & c + d + e \\ c & d & a + b + e \\ b + d + e & a + c + e & -e \end{pmatrix}.$$

Conversely, each of its six sums is $a + b + c + d + e$. The five coefficient matrices are

$$M_a = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad M_b = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad M_c = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

$$M_d = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad M_e = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & -1 \end{pmatrix}.$$

They span the family. In a zero combination, entries $(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)$ give the five coefficients as zero. They are a basis; the dimension is 5.

Exercise 3.8 (Extracting a basis with prescribed members). Let

$$A = \begin{pmatrix} 0 & 1 & 0 & -3 & 2 \\ 0 & -2 & 0 & 6 & 2 \\ 0 & 2 & 2 & 5 & 2 \\ 0 & -4 & -2 & 1 & 0 \end{pmatrix}, \quad u = \begin{pmatrix} 2 \\ -1 \\ 3 \\ -4 \end{pmatrix}, \quad v = \begin{pmatrix} 0 \\ 9 \\ -3 \\ 12 \end{pmatrix}.$$

- Express u, v as combinations of the columns of A .
- Extract a basis of $\mathcal{C}(A)$ by scanning the columns in order.
- Find rank A .
- Find a basis of $\mathcal{C}(A)$ that includes u, v .

Solution. Write a_j for column j . (a) Direct comparison gives $u = a_2 + \frac{1}{2}a_5$, $v = -3a_2 + \frac{3}{2}a_5$.

(b) Discard $a_1 = 0$, retain a_2, a_3 , and discard $a_4 = -3a_2 + \frac{11}{2}a_3$. The column a_5 must be retained. In fact, if $\alpha a_2 + \beta a_3 + \gamma a_5 = 0$, the first two coordinates give $\alpha + 2\gamma = 0$, $-2\alpha + 2\gamma = 0$, so $\gamma = \alpha = 0$; the third gives $2\beta = 0$. Thus (a_2, a_3, a_5) is the extracted basis.

(c) This basis has three members, hence rank $A = 3$.

(d) In a relation $su + tv + wa_3 = 0$, substitute (a):

$$(s - 3t)a_2 + wa_3 + (s/2 + 3t/2)a_5 = 0.$$

Part (b) gives $s = 3t$, $w = 0$, $s + 3t = 0$; hence $s = t = w = 0$. The three independent members of the three-dimensional space form a basis. For an explicit spanning check, $a_2 = u/2 - v/6$ and $a_5 = u + v/3$, so every original column lies in $\text{span}(u, v, a_3)$.

- Exercise 3.9** (Basis exchange). (a) If $(x_1, \dots, x_p)$ and $(y_1, \dots, y_q)$ are independent lists in $\mathbb{R}^n$, with $p < q$, show that some y_i can be adjoined to the first list while preserving independence.
- (b) Let X, Y be bases of $\mathbb{R}^n$. If $S \subseteq X$ contains r members, show that there is a subset $T \subseteq Y$ of r members for which $(X \setminus S) \cup T$ is a basis.

Solution. (a) If no y_i can be adjoined, Proposition 3.1 places every y_i in $\text{span}(x_1, \dots, x_p)$. This gives $q > p$ independent vectors in a space spanned by p vectors, contradicting Theorem 3.2.

(b) Start with the $n - r$ independent members of $X \setminus S$. While the current list has fewer than n members, apply (a) with the n -member basis Y , adjoining one of its members outside the current span. No member can be chosen twice. After r additions the list is independent of length n , hence a basis. The added members form the required T . For $r = 0$, use $T = \emptyset$.

Exercise 3.10 (A product space). For subspaces $U \subseteq \mathbb{R}^m$ and $W \subseteq \mathbb{R}^n$, define

$$U \times W = \left\{ \begin{pmatrix} u \\ w \end{pmatrix} : u \in U, w \in W \right\}.$$

Given bases $(u_1, \dots, u_r)$, $(w_1, \dots, w_s)$, find a basis of $U \times W$ and prove $\dim(U \times W) = r + s$.

Solution. The set is a subspace of $\mathbb{R}^{m+n}$ by componentwise closure. A basis is

$$\begin{pmatrix} u_1 \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} u_r \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ w_1 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ w_s \end{pmatrix}.$$

It spans because the two components can be expanded separately. In a zero combination, the upper block gives $\sum_i a_i u_i = 0$, and the lower block gives $\sum_j b_j w_j = 0$. Both original lists are independent, so all coefficients vanish. The basis has $r + s$ members.

Rank, factorization, and solution sets

Exercise 3.11 (An exceptional parameter). Find the rank, for every $\alpha \in \mathbb{R}$, of

$$A_\alpha = \begin{pmatrix} 1 & 4 & \alpha & 4 \\ 2 & -6 & 7 & 1 \\ 3 & 2 & -6 & 7 \\ 2 & 2 & -5 & 5 \end{pmatrix}.$$

Solution. Subtract $2R_1, 3R_1, 2R_1$ from rows 2, 3, 4, respectively:

$$\begin{pmatrix} 1 & 4 & \alpha & 4 \\ 0 & -14 & 7 - 2\alpha & -7 \\ 0 & -10 & -6 - 3\alpha & -5 \\ 0 & -6 & -5 - 2\alpha & -3 \end{pmatrix}.$$

Next replace R_3 by $7R_3 - 5R_2$, R_4 by $7R_4 - 3R_2$, and then R_4 by $11R_4 - 8R_3$, obtaining

$$\begin{pmatrix} 1 & 4 & \alpha & 4 \\ 0 & -14 & 7 - 2\alpha & -7 \\ 0 & 0 & -11(\alpha + 7) & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Each replacement is a nonzero constant scaling followed by a row addition, so it is valid for every α . The first two pivots never vanish. Therefore $\text{rank } A_\alpha = 2$ for $\alpha = -7$, and 3 otherwise.

Exercise 3.12 (Two rank factorizations). Find two different rank factorizations of

$$A = \begin{pmatrix} 1 & 1 & 2 & 0 \\ 2 & -3 & 1 & 1 \\ 3 & -2 & 3 & 1 \\ 5 & -5 & 4 & 2 \end{pmatrix}.$$

Solution. The first two columns a_1, a_2 are independent: a zero combination gives $s + t = 0$, $2s - 3t = 0$, hence $s = t = 0$. The remaining columns satisfy $a_3 = (7a_1 + 3a_2)/5$, $a_4 = (a_1 - a_2)/5$. Thus

$$A = \underbrace{\begin{pmatrix} 1 & 1 \\ 2 & -3 \\ 3 & -2 \\ 5 & -5 \end{pmatrix}}_B \underbrace{\begin{pmatrix} 1 & 0 & 7/5 & 1/5 \\ 0 & 1 & 3/5 & -1/5 \end{pmatrix}}_F.$$

The factors have rank 2, so this is a rank factorization. Choose a_1, a_4 instead. Since $a_2 = a_1 - 5a_4$ and $a_3 = 2a_1 - 3a_4$,

$$A = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 1 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 2 & 0 \\ 0 & -5 & -3 & 1 \end{pmatrix}.$$

The first factor has independent columns by its first two rows; columns 1 and 4 of the second factor form I_2 , proving its rank is 2.

Exercise 3.13 (A rectangular rank calculation). Find the rank of

$$B = \begin{pmatrix} 1 & 5 & 6 & 3 \\ 2 & 1 & 4 & 1 \\ 3 & 5 & 5 & 4 \end{pmatrix}.$$

Can any 3×4 matrix have rank 4?

Solution. Use $R_2 \leftarrow R_2 - 2R_1$, $R_3 \leftarrow R_3 - 3R_1$, and then $R_3 \leftarrow 9R_3 - 10R_2$:

$$B \longrightarrow \begin{pmatrix} 1 & 5 & 6 & 3 \\ 0 & -9 & -8 & -5 \\ 0 & -10 & -13 & -5 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 5 & 6 & 3 \\ 0 & -9 & -8 & -5 \\ 0 & 0 & -37 & 5 \end{pmatrix}.$$

There are three pivots, so $\text{rank } B = 3$. Every 3×4 matrix has its column space inside $\mathbb{R}^3$, so its rank cannot exceed 3.

Exercise 3.14 (Diagonal entries and rank). Prove that a diagonal matrix has rank equal to the number of its nonzero diagonal entries. Does the same statement hold for triangular matrices?

Solution. The nonzero columns of a diagonal matrix are nonzero multiples of distinct standard columns, hence independent; all other columns are zero. For triangular matrices, consider

$$L = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

The first has rank 1 and no nonzero diagonal entry. The second has rank 2, from its first two independent columns, but only one nonzero diagonal entry.

Exercise 3.15 (Rank one). Show that a real $m \times n$ matrix A has rank 1 if and only if $A = ab^\top$ for some nonzero $a \in \mathbb{R}^m$, $b \in \mathbb{R}^n$.

Solution. If $A = ab^\top$, its columns are $b_j a$. They lie in $\text{span}(a)$, and some $b_j \neq 0$, so one column spans that one-dimensional space. Conversely, if $\text{rank } A = 1$, choose a nonzero column a . It is a basis of $\mathcal{C}(A)$, so each column is $b_j a$. Collecting these coefficients gives $A = ab^\top$; the coefficient of the chosen column is 1, ensuring $b \neq 0$.

Exercise 3.16 (Rank as a number of outer products). If $\text{rank } A = r$, express A as the sum of r rank-one matrices. Also show that fewer than r such summands cannot suffice.

Solution. For $r > 0$, take a rank factorization $A = BF$, write the columns of B as b_i and the rows of F as $f_i^\top$, and expand:

$$A = \sum_{i=1}^r b_i f_i^\top.$$

The columns of B and rows of F are independent, so none is zero. Exercise 3.15 makes each summand rank one. If instead $A = \sum_{i=1}^k u_i v_i^\top$, every column of A is a combination of $u_1, \dots, u_k$, whence $\text{rank } A \leq k$. Thus $k \geq r$. For $A = 0$, the empty sum has zero summands.

Exercise 3.17 (* Lifting an image basis). Let $x_1, \dots, x_r$ be a basis of $\mathcal{C}(A)$, and $y_1, \dots, y_s$ a basis of $N(A)$, where $A : m \times n$. For each j , choose z_j with $Az_j = x_j$. Prove directly that $(y_1, \dots, y_s, z_1, \dots, z_r)$ is a basis of $\mathbb{R}^n$, and deduce $r + s = n$.

Solution.

- For $v \in \mathbb{R}^n$, expand $Av = \sum_j a_j x_j$. Then

$$A \left(v - \sum_j a_j z_j \right) = Av - \sum_j a_j x_j = 0,$$

so $v - \sum_j a_j z_j = \sum_i b_i y_i$ for suitable b_i . Thus $v = \sum_i b_i y_i + \sum_j a_j z_j$, proving spanning.

- In a zero combination $\sum_i b_i y_i + \sum_j a_j z_j = 0$, apply A to obtain $\sum_j a_j x_j = 0$. Independence of the x_j gives all $a_j = 0$; independence of the y_i then gives all $b_i = 0$.
- The combined list is a basis, so its length is n . The argument includes empty lists when $r = 0$ or $s = 0$.

Exercise 3.18 (An additional column). Suppose $A : m \times n$ and there is $a \in \mathbb{R}^m$ such that $\text{rank}[A \ a] = \text{rank } A + 1$. Prove that the rows of A are dependent.

Solution. There are only m rows in the augmented matrix, so $\text{rank } A + 1 \leq m$. Thus $\text{rank } A < m$, and its m rows cannot be independent.

Exercise 3.19 (Ranks of sums and differences). For matrices A, B of the same size, prove

$$\text{rank}(A + B) \leq \text{rank } A + \text{rank } B, \quad \text{rank}(A - B) \geq |\text{rank } A - \text{rank } B|.$$

Solution. Every column of $A + B$ belongs to $\mathcal{C}(A) + \mathcal{C}(B)$. By the dimension formula,

$$\text{rank}(A + B) \leq \dim(\mathcal{C}(A) + \mathcal{C}(B)) \leq \text{rank } A + \text{rank } B.$$

Apply this result to $A = (A - B) + B$ to obtain $\text{rank}(A - B) \geq \text{rank } A - \text{rank } B$. Interchange A, B , noting that multiplication by -1 leaves a span unchanged, to get the other lower bound.

Exercise 3.20 (Deleting rows and columns). If A_1 is a submatrix of A , prove $\text{rank } A_1 \leq \text{rank } A$.

Solution. Delete the unwanted rows first, obtaining B . Its row space is spanned by a subset of the rows of A , so $\text{rank } B \leq \text{rank } A$. Now delete the unwanted columns of B . The remaining columns span a subspace of $\mathcal{C}(B)$, so $\text{rank } A_1 \leq \text{rank } B$. Combining the inequalities proves the result.

Block structure and rank

Exercise 3.21 (Two zero blocks). For arbitrary rectangular blocks, with zero blocks of the required sizes, prove

$$\text{rank} \begin{pmatrix} A & 0 \\ 0 & D \end{pmatrix} = \text{rank } A + \text{rank } D, \quad \text{rank} \begin{pmatrix} 0 & B \\ C & 0 \end{pmatrix} = \text{rank } B + \text{rank } C.$$

Solution. Take bases $a_1, \dots, a_r$ of $\mathcal{C}(A)$ and $d_1, \dots, d_s$ of $\mathcal{C}(D)$. The columns

$$\begin{pmatrix} a_i \\ 0 \end{pmatrix} \quad (1 \leq i \leq r), \quad \begin{pmatrix} 0 \\ d_j \end{pmatrix} \quad (1 \leq j \leq s)$$

span the block matrix's column space. A zero combination has upper block $\sum_i c_i a_i = 0$ and lower block $\sum_j e_j d_j = 0$, so all coefficients vanish. Its rank is $r + s$. For the second identity, interchange the two column groups to obtain $\begin{pmatrix} B & 0 \\ 0 & C \end{pmatrix}$. A reordering of columns does not change their span, so the first identity applies.

Exercise 3.22 (A single zero block is insufficient). Show that neither of the identities

$$\text{rank} \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \text{rank } A + \text{rank } D, \quad \text{rank} \begin{pmatrix} A & 0 \\ C & D \end{pmatrix} = \text{rank } A + \text{rank } D$$

holds in general.

Solution. Take $A = D = [0]$ and $B = C = [1]$. Each block matrix has one nonzero column and rank 1, whereas $\text{rank } A + \text{rank } D = 0$. More generally, if the diagonal blocks vanish, the respective ranks are $\text{rank } B$ and $\text{rank } C$.

Exercise 3.23 (A bordered matrix). For $A \in \mathbb{R}^{m \times n}$, $a \in \mathbb{R}^n$, and $\alpha \in \mathbb{R}$, let

$$Z = \begin{pmatrix} 0 & A \\ \alpha & a^\top \end{pmatrix}.$$

Prove

$$\text{rank } Z = \begin{cases} \text{rank } A, & \alpha = 0 \text{ and } a \in \mathcal{R}(A), \\ \text{rank } A + 1, & \text{otherwise.} \end{cases}$$

Solution. The transposed first m rows span

$$U = \left\{ \begin{pmatrix} 0 \\ r \end{pmatrix} : r \in \mathcal{R}(A) \right\}.$$

If $(r_1, \dots, r_k)$ is a basis of $\mathcal{R}(A)$, the columns $\begin{pmatrix} 0 \\ r_i \end{pmatrix}$ form a basis of U : their lower blocks prove both spanning and independence. Thus $\dim U = \text{rank } A$. The transposed last row is $z = \begin{pmatrix} \alpha \\ a \end{pmatrix}$, and

$$z \in U \iff \alpha = 0 \text{ and } a \in \mathcal{R}(A).$$

Adjoining z leaves the dimension unchanged in this case and increases it by one otherwise, by Proposition 3.1.

Exercise 3.24 (Equivalence of rectangular matrices). For $A, B \in \mathbb{R}^{m \times n}$, prove that

$$B = PAQ \text{ for some invertible } P, Q \iff \text{rank } A = \text{rank } B.$$

Solution.

- The forward direction is Proposition 3.9.
- For the converse, first prove that every rank- r matrix A can be changed to $J_r = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$ by invertible multiplication. For $r > 0$, take $A = CF$ with full rank factors. Extend the columns of C to a basis of $\mathbb{R}^m$, forming an invertible matrix M . Extend the transposes of the rows of F to a basis of $\mathbb{R}^n$; use their transposes as the rows of an invertible matrix N . Invertibility follows from the square criterion in Unit 2 and equality of row and column ranks. Then

$$M^{-1}C = \begin{pmatrix} I_r \\ 0 \end{pmatrix}, \quad FN^{-1} = [I_r \ 0], \quad M^{-1}AN^{-1} = J_r.$$

For $r = 0$, $A = 0 = J_0$, so identity multipliers suffice.

- Apply this construction to A, B of the same rank: $P_1AQ_1 = J_r = P_2BQ_2$. Therefore

$$B = P_2^{-1}P_1AQ_1Q_2^{-1}.$$

Both new multipliers are invertible by the product inverse law.

Exercise 3.25 (Stacking rows). For $A : m \times n$, $C : q \times n$, and $D = \begin{pmatrix} A \\ C \end{pmatrix}$, prove

- $\mathcal{R}(D) = \mathcal{R}(C)$ if and only if $\mathcal{R}(A) \subseteq \mathcal{R}(C)$.
- $\text{rank } D \geq \text{rank } C$, with equality exactly under the same inclusion.

Solution. The rows of D consist of the two lists of rows, so $\mathcal{R}(D) = \mathcal{R}(A) + \mathcal{R}(C)$. This equals $\mathcal{R}(C)$ exactly when $\mathcal{R}(A) \subseteq \mathcal{R}(C)$, proving (a). The inclusion $\mathcal{R}(C) \subseteq \mathcal{R}(D)$ gives (b)'s inequality. For nested finite-dimensional subspaces, equal dimensions are equivalent to equal spaces, proving the equality criterion.

Exercise 3.26 (Changing a few entries). If B is obtained from $A \in \mathbb{R}^{m \times n}$ by changing k entries, prove

$$\text{rank } A - k \leq \text{rank } B \leq \text{rank } A + k.$$

Solution. Let $D = B - A$. At most k entries of D are nonzero, so at most k rows are nonzero; its row space is spanned by at most k vectors. Thus $\text{rank } D \leq k$. By Exercise 3.19,

$$\text{rank } B = \text{rank}(A + D) \leq \text{rank } A + k, \quad \text{rank } A = \text{rank}(B - D) \leq \text{rank } B + k.$$

Rearranging the second inequality gives the lower bound.